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Routing Protocols in the Wild

RIP, OSPF, BGP & AWS Route Tables

CS Computer Networks · Week 9 · Lecture 18 of 24

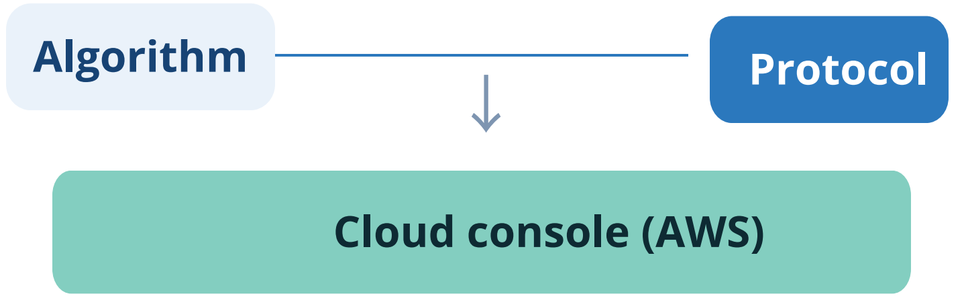
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Recap: From Algorithm to Protocol

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Where we are in the story

- Last lecture gave us the **algorithms**: Bellman-Ford and Dijkstra.
- Real routers run them as **Named protocols:** with knobs, metrics and limits.
- And the cloud hides all of it behind a console you just fill in.



Same job: Move packets to the right place, three very different ways to say so.

Warm-Up: Fewer Hops ≠ Faster

A trap of counting only hops

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- A protocol that measures distance **only in hops** picks the 1-hop satellite link.
- But satellite is slow; the 2-hop fibre path is far faster and cheaper.
- **Analogy: choosing the road with the fewest turns, even if it's gravel.**

1 hop · satellite
600 ms latency — RIP picks this

2 hops · fibre
8 ms latency — actually better

Which link would you choose?

If the metric is wrong, the "best" path is wrong. This motivates OSPF.

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Two Worlds: Interior vs Exterior

Streets inside your city vs highways between cities.

Inside vs Between Networks

The single line that organises this whole lecture

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Interior routing

within one organisation

- Goal: find the best path
- Protocols: RIP, OSPF
- Everyone trusts everyone

Exterior routing

between organisations

- Goal: honour business policy
- Protocol: BGP
- Nobody fully trusts anybody

Inside your building you optimise for speed; between buildings you negotiate.

RIP: Just Count the Hops

Distance-vector, the simplest of the three

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- Metric = hop count; **max 15**, and 16 means "unreachable".
- Periodic vector exchange — Bellman-Ford in practice.
- **Pro: trivial to configure. Good for small networks only.**
- Cons: slow convergence, count-to-infinity, ignores link speed.

"How far?"

3 hops

Doesn't ask "how fast" or "how wide" — only "how many routers away".

Counting Doors, Not Distance

Like judging a journey by the number of towns you pass, not the miles.

OSPF — See the Whole Map

Link-state routing at enterprise scale

- Cost metric (often bandwidth-based) — reflects link quality.
- Dijkstra under the hood; every router builds the full topology.
- **Areas give hierarchy so it scales to large networks.**
- Fast convergence via link-state flooding — the enterprise default.

Path A · 2 hops · gravel	cost 100
Path B · 3 hops · fibre	cost 12 ✓

OSPF picks Path B — more hops, lower cost.

Weighing Roads, Not Turns

A GPS that knows road speeds beats one that only counts intersections.

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RIP vs OSPF at a Glance

Property	RIP	OSPF
Type	Distance-vector	Link-state
Metric	Hop count (max 15)	Cost (bandwidth)
Convergence	Slow	Fast (flooding)
Best for	Tiny networks	Large enterprises

Checkpoint: why does OSPF choose better paths? Its metric reflects link quality, RIP only counts hops.

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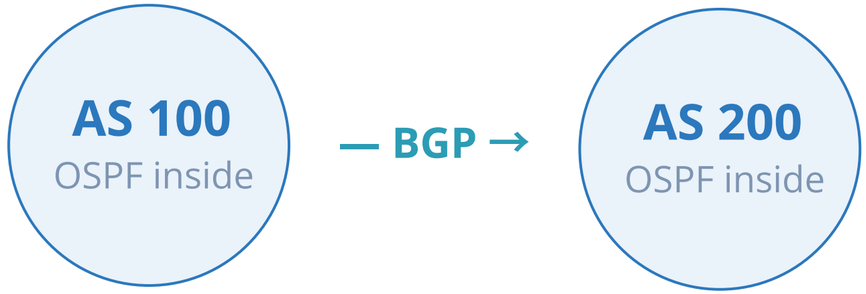
BGP: The Glue Between Networks

Where routing becomes politics and economics.

Autonomous Systems & Path-Vector

Countries with their own internal rules

- Each ISP / large network is an **Autonomous System (AS)**.
- OSPF / RIP run **inside** each AS; BGP runs **between** them.
- BGP advertises full **paths** — the sequence of AS numbers to a destination.
- Callback: CIDR aggregation (L13) keeps these BGP tables manageable.




```
path: AS100 → AS200 → AS300
```

Interior Inside, BGP Between

Inside a country you drive freely; crossing borders you follow treaties.

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Policy Beats Shortest Path

Why the Internet's routing is a business decision

- BGP chooses by **business policy**, not just the shortest path.
- An ISP may prefer a **customer route** over a cheaper transit path.
- Two ISPs can pick different paths to the **same** destination.
- Understand BGP's **role** — you won't configure it here.

- Customer route — **preferred** (earns revenue)
- Transit route — cheaper hops, but rejected

Follow the Money, Not the Miles

The globe doesn't run one giant Dijkstra — it runs thousands of negotiated deals.

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The Cloud Abstraction

You state where traffic goes; AWS makes it happen.

AWS Route Tables — Just Declare It

No RIP, no OSPF — you fill in a table

- In a VPC you **declare** routes; AWS implements them for you.
- Destination + target : each row is one forwarding rule.
- **Declarative: you state intent, not the negotiation.**
- Callback: same VPC constructs from Lecture 15.

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-0a1b
172.16.0.0/16	vgw-9f8e

A VPC Route Table

Like a GPS: you type the destination; you don't teach it every road yourself.

Where BGP Sneaks Back In

The hybrid edge between on-prem and AWS



Two Worlds Shake Hands

- Inside the VPC everything stays declarative.
- **Site-to-Site VPN and Direct Connect use BGP** at the edge.
- Route propagation dynamically learns on-prem routes into the table.
- Full hybrid story arrives in Lecture 24.

Checkpoint: where does BGP show up in a console-managed AWS network?

Declarative vs Distributed

Same forwarding outcome, very different mechanism

Traditional · distributed

- Routers talk and negotiate
- Tables emerge from the protocol
- Packet Tracer OSPF table

Cloud · declarative

- You state intent in a table
- AWS enforces it under the hood
- VPC route table

Both tables end at the same forwarding decision : one negotiated, one declared.

✦ HIGHLIGHT · LIVE DEMO

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The Packet Tracer Bake-Off

1 · Static routes

Configure static routes — it works. Then down a link: **traffic fails.** Static routes don't adapt.

2 · Enable OSPF

Drop the same link — the network **reconverges and heals in seconds.** The payoff of dynamic routing.

3 · AWS route table

Static-route **simplicity**, with the dynamic parts **managed for you.**

Watch the network heal itself, then let the cloud do the healing for you.

Quick Checkpoints

Q1 · Which protocol runs between autonomous systems?

a) RIP | b) OSPF | **c) BGP ✓** | d) ICMP

Q2 · True or False: "In AWS you typically run OSPF inside your VPC."

False — VPCs use declarative route tables, not a routing protocol.

Interior vs exterior, and declarative vs distributed, the two axes to remember.

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Activity: Pick the Protocol

Match each scenario to the right approach

Tiny LAN, a few routers	**RIP / static**
ISP connecting to another ISP	**BGP**
Large multi-building campus	**OSPF**
Traffic inside an AWS VPC	**Route table**

The scale and the boundary decide the tool, not personal preference

Please fill the feedback form.

Thanks for watching!